Harsh Vibhuti

hv9614@gmail.com | St. Louis, MO | LinkedIn | Portfolio

Mechanical Engineering and Mathematics student skilled in SolidWorks, structural FEA, thermal CFD, GD&T, DFM/DFA, CNC machining, and electromechanical prototyping. Excited to apply my skills to hands on engineering design and real world challenges

EDUCATION

Washington University in St. Louis | St. Louis, MO *Expected: May 2027*

B.S. Mechanical Engineering and Mathematics | Minor: Robotics | GPA: 3.8

TECHNICAL SKILLS

Design and Analysis: SolidWorks (CSWP certified), 3D Modeling, ANSYS Fluent, Icepak, COMSOL, FEA, Tolerance Stack-up, Enclosures, Chassis, Brackets, Trays, Packaging, Sheet Metal, Plastic and Die-Cast Components, Part & Assembly Modeling, Hardware Validation, Vibrations & Environmental Testing, SLA/FDM 3D Printing

Manufacturing: DFM/DFA, GD&T, 2D Drawings, FeatureCAM, CNC Machining (3-axis and 4-axis), Injection Molding, Stamping, PLM/PDM, BOM Creation, Prototyping, ECO, Failure Analysis, RCCA, Mechanical Tools, Supply Chain & Vendor Coordination, Technical Documentation

Electromechanical: PCB Layouts, Thermal Optimization, Cable Routing, Rack Integration, Servo & Actuator Integration, EMI/EMC & Safety Compliance, Python (NumPy, Pandas, SciPy), MATLAB, HTML, C++, Data Center Hardware

TECHNICAL EXPERIENCE

WashU Rocketry | Mechanical Design and Test Engineer *St. Louis, MO | Sep 2025 - Present*

- Conducted a Delrin vs. aluminum tradeoff study for rocket bulkheads designed in SolidWorks applying GD&T principles and achieved 44% mass reduction and 4x strength improvement using iterative FEA and conducted fit checks, vibration tests, and tolerance stackups.
- Created assembly for rocket, 2D drawings, and BOM. Sourced hardware from vendors, programmed a 3-axis CNC toolpath in FeatureCAM, and led CNC milling for the NASA USLI competition.
- Designed, SLA-printed, and validated custom resin bearings to replace commercial ball casters, executing prototype testing and failure analysis to reduce mechanism mass, rolling friction, and unit cost.

WashU Design Build Fly | Avionics System Co-lead St. Louis, MO | *Sep 2025 - Present*

- Designed and 3D printed a modular avionics enclosure, mounting trays, and custom servo brackets in SolidWorks, making use of DFM/DFA principles to optimize internal packaging and resolve service-access constraints for rapid troubleshooting.
- Leading a 12-member interdisciplinary team to integrate electromechanical assemblies, actuators, and control linkages with structured cable routing and harness management, reducing troubleshooting time by 2x.

Capital Area Council of Governments | Air Quality Intern *Austin, TX | May 2025 - Aug 2025*

- Automated the collection of Texas-wide air quality data with the use of Python scripts, reducing QA time by 95%.
- Presented the statistical data findings as policy and operational recommendations for 200+ regional stakeholders, municipalities, and directors of the Clean Air Coalition of Central Texas.

ENGINEERING PROJECTS

Atomization Chamber | Mechanical Design Project, Dr. Friend Lab *St. Louis, MO | Aug 2026 - Present*

- Designed a 40 mm × 40 mm methanol atomization enclosure in SolidWorks, increasing reusability 10× and improving sealing performance 400x based on customer needs.
- Conducted CFD and thermal simulations using ANSYS Fluent and Icepak to evaluate carrier-gas flow and heat transfer, targeting 7× fuel-cell efficiency over liquid-methanol delivery.
- Applied GD&T and tolerance stack-ups to threaded interfaces and O-ring seals; created the BOM, fabricated SLA/FDM prototypes, and ran pressure leak tests with failure analysis and RCCA.

Autonomous LLM Robot | Personal Robotics Project *St. Louis, MO | Aug 2026 - Present*

- Built an autonomous mobile robot for people with disability, packaging Arduino control boards, IR sensors, geared DC actuators, motor-driver circuitry, and battery power with structured cable routing on a chassis.
- Architected a Python-to-Arduino serial interface integrating speech-to-text recognition and LLM command parsing to execute natural-language tasks, designing the system for Raspberry Pi edge deployment.

LEADERSHIP

WashU Engineers Without Borders | Vice President *St. Louis, MO | Jan 2026 - Present*

- Lead cross-functional design reviews, technical communication, and project coordination for a clean-water distribution project serving a primary school in Uganda.